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ECE 240A

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ECE 240A Final Design Project

1. Overlap between resonator mode and pump and coupling optics

To simplify calculations, we want to minimize the effects of the Gaussian profile of the pump and approximate the profiles as uniform. This is further simplified by adopting the assumption of homogenous broadening and the laser oscillating on only one cavity mode so that all the atoms behave in the same manner. To do this, we determine the value of the focal length f and L_1 distance of the coupling optic from the pump that ensures that the wavefront returns an infinite output beam radius of curvature. The ABCD matrix gives us

$$\frac{1}{R_2} = \frac{1}{R_1} - \frac{1}{f}$$

where R_2 is the radius of curvature of the outgoing beam from the coupling optic and R_1 is the incoming beam radius of curvature. We have the following equations

$$R(z) = z\left[1 + \left(\frac{z_0}{z}\right)^2\right]$$

$$z_0 = \frac{\pi\omega_0^2}{\lambda}$$

to find the radius of curvature of the beam at any point z from the minimum beam waist. Here we will assume that $z = 0$ occurs right at the pump and provide a value of $\omega_0 = 175\ \mu m$ to the minimum beam waist.

$$z_0 = \frac{\pi\omega_0^2}{\lambda} = \frac{\pi(150 * 10^{-6})^2}{(808.5 * 10^{-9})}$$

In order for R_2 to be infinite, $R_1 = f$

$$\frac{1}{R_2} = \frac{1}{R_1} - \frac{1}{f} \rightarrow \frac{1}{R_2} = \frac{1}{f} - \frac{1}{f} \rightarrow \frac{1}{R_2} = 0 \rightarrow R_2 = \infty$$

$$R(L_1) = L_1\left[1 + \left(\frac{z_0}{L_1}\right)^2\right]$$

$$f = L_1 + \frac{z_0^2}{L_1^2} = \frac{L_1^2 + z_0^2}{L_1}$$

$$fL_1 = L_1^2 + z_0^2$$

$$0 = L_1^2 - L_1 f + z_0^2$$

We plug this function into MATLAB and range the values of f to be between 0 and 1 meter to find the lowest L_1 value between 2 and 5 cm that still achieves a collimated beam. Thus, we have

$L_1 = 2.01\ cm$

$f = 40\ cm$

From this, we can determine what the beam waist radius will be throughout the laser system since the beam waist radius at the coupling lens will be equivalent to the beam waist radius exiting the lens.

$$\omega^2(z) = \omega_0^2\left[1 + \left(\frac{z}{z_0}\right)^2\right]$$

$$\omega^2(2.01 * 10^{-2}) = (150 * 10^{-6})^2\left[1 + \left(\frac{808.5 * 10^{-9} * 2.01 * 10^{-2}}{\pi(150 * 10^{-6})^2}\right)^2\right]$$

$\omega = 154\ \mu m$

$$\%\ diff = \frac{154 - 150}{150} * 100 > 2\%$$

Since our initial assumption for a beam waist at 150 μm did not satisfy the 2% criteria for considering the beam uniform, we increase the beam waist to 175 μm and redo the calculations.

$L_1 = 2.09\ cm$

$f = 70\ cm$

$\omega = 178\ \mu m$

$$\%\ diff = \frac{178 - 175}{175} * 100 = 1.5\% < 2\%$$

We verify, first, that this beam waist we calculated meets the 1 cm transverse constraint as $2 * 178\ \mu m \ll 1\ cm$. Next, we find the diameter of the thin lens that allows 99.9% of light intensity through based on the value of ω_0 to ensure that the pump beam is not clipped and thus allows us to treat it as an infinite transverse beam.

$$\frac{1}{\pi\omega_0^2}\int_0^{2\pi}\int_0^{d/2} e^{-\frac{r^2}{\omega_0^2}} r dr d\phi = 0.999$$

$$\frac{1}{\pi\omega_0^2}\int_0^{d/2} e^{-\frac{r^2}{\omega_0^2}} r dr \int_0^{2\pi} d\phi = 0.999$$

$$2\pi * \frac{1}{\pi\omega_0^2}\int_0^{\frac{d}{2}} e^{-\frac{r^2}{\omega_0^2}} rdr = 0.999$$

$$u = \frac{r}{\omega_0}$$

$$du = \frac{1}{\omega_0}dr$$

$$2\int_0^{\frac{d}{2\omega_0}} ue^{-u^2}\, du = 0.999$$

$$2\left[-\frac{1}{2}ue^{-u^2}\right] = 0.999$$

$$-ue^{-u^2} = 0.999$$

$$1 - e^{-\left(\frac{d}{2\omega_0}\right)^2} = 0.999$$

$$-e^{-\left(\frac{d}{2\omega_0}\right)^2} = 0.999 - 1$$

$$e^{-\left(\frac{d}{2\omega_0}\right)^2} = 0.001$$

$$\ln e^{-\left(\frac{d}{2\omega_0}\right)^2} = \ln 0.001$$

$$-\left(\frac{d}{2\omega_0}\right)^2 = \ln 0.001$$

$$\left(\frac{d}{2\omega_0}\right)^2 = -\ln 0.001$$

$$\frac{d}{2\omega_0} = \sqrt{-\ln 0.001}$$

$$d = 2\omega_0\sqrt{-\ln 0.001}$$

$$d = 2(175 * 10^{-6})\sqrt{-\ln 0.001}$$

$$d = 919.89\ \mu m\ \approx 1\ mm$$

We defined the back mirror of the resonator to have an infinite radius and is thus flat. This simplifies our calculations and allows us to ignore the effects of the back reflector acting as a lens that influences the beam profile as the beam continues traversing collimated.

The overlap between the resonator mode and pump and coupling optics will be described by profiling the intensity of the beam before and after it enters the gain medium. We are given the pump power $P_0 = 1\ W$ which we use to solve for the pump intensity:

$$I_0 = \frac{2P_0}{\pi\omega_0^2} = \frac{2(1)}{\pi(175 * 10^{-4})^2} = 2078.8\frac{W}{cm^2}$$

Then, we can solve for the intensity of the beam at the center exiting the coupling optic.

$$I(r,z) = I_0\left(\frac{\omega_0}{\omega(z)}\right)^2\left[e^{-\frac{2r^2}{\omega^2(z)}}\right]$$

$$I(0,L_1) = 2078.8\left(\frac{175}{178}\right)^2\ [e^0] = 2016.8\frac{W}{cm^2}$$

Since we assume a uniform beam profile following the coupling optic into the cavity, we continue with the assumption that the beam intensity is also uniform. The only thing we need to keep into account is that the transmittance of the back mirror into the cavity is 0.95. Thus, we define the input intensity into the gain medium to be

$$I_1 = 0.95 * I(0,L_1) = 0.95 * 2016.8 = 1916\frac{W}{cm^2}$$

The power of the beam at the doubler P_2 multiplied by the conversion efficiency L will result in the output power of the green wavelength light. We first analyze what the power of the beam is entering the gain medium.

$$P_1 = I_1 A_{beam} = 1916 * \frac{\pi(178 * 10^{-4})^2}{2} = 0.95\ W$$

We design our laser pointer with the goal to output no more than $P_{out} = 1\ mW$ of power for the green wavelength light. This requires us to determine what the small signal power gain G_0 and the output power from the gain medium P_2 is. This intensity value at the doubler represents the average intensity of the beam calculated over a circular area with a radius equal to the waist of the beam at the doubler.

$$P_{out} = 1000 * L * P_2$$

$$I_2 = \frac{P_2}{A} = \frac{2P_2}{\pi\omega(z)^2}$$

$$L = \tanh^2\left(\sqrt{I_2}\right) = \tanh^2\left(\sqrt{\frac{2P_2 * 10^{-9}}{\pi\omega(z)^2}}\right) = \tanh^2\left(\sqrt{\frac{2P_2 * 10^{-9}}{\pi(178 * 10^{-4})^2}}\right)$$

$$1\ mW = 1000 * \tanh^2\left(\sqrt{\frac{2P_2 * 10^{-9}}{\pi(178 * 10^{-4})^2}}\right) * P_2 * 10^{-9}$$

We use MATLAB to solve the above formula for P_2 and get

$P_2 = 22.2668\ W$

$$L = \tanh^2\left(\sqrt{\frac{2 * 22.2668\ * 10^{-9}}{\pi(178 * 10^{-4})^2}}\right) = 0.0000449072$$

These values allow us to determine our small signal power gain

$$G_0 = \frac{P_2}{P_1} = \frac{22.2668}{0.95} = 23.44\ \approx 24$$

Since we have a completely flat mirror composing one end of the resonator, we now consider what the radius R_2 of the other mirror needs to be to maintain our assumption of a uniform beam profile.

$$T = \begin{bmatrix} 1 & d + z_1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -\frac{1}{f} & 1 \end{bmatrix} \begin{bmatrix} 1 & d - z_1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 - \frac{d + z_1}{f} & d + z_1 + (d - z_1)\left(1 - \frac{d + z_1}{f}\right) \\ -\frac{1}{f} & 1 - \frac{d - z_1}{f} \end{bmatrix}$$

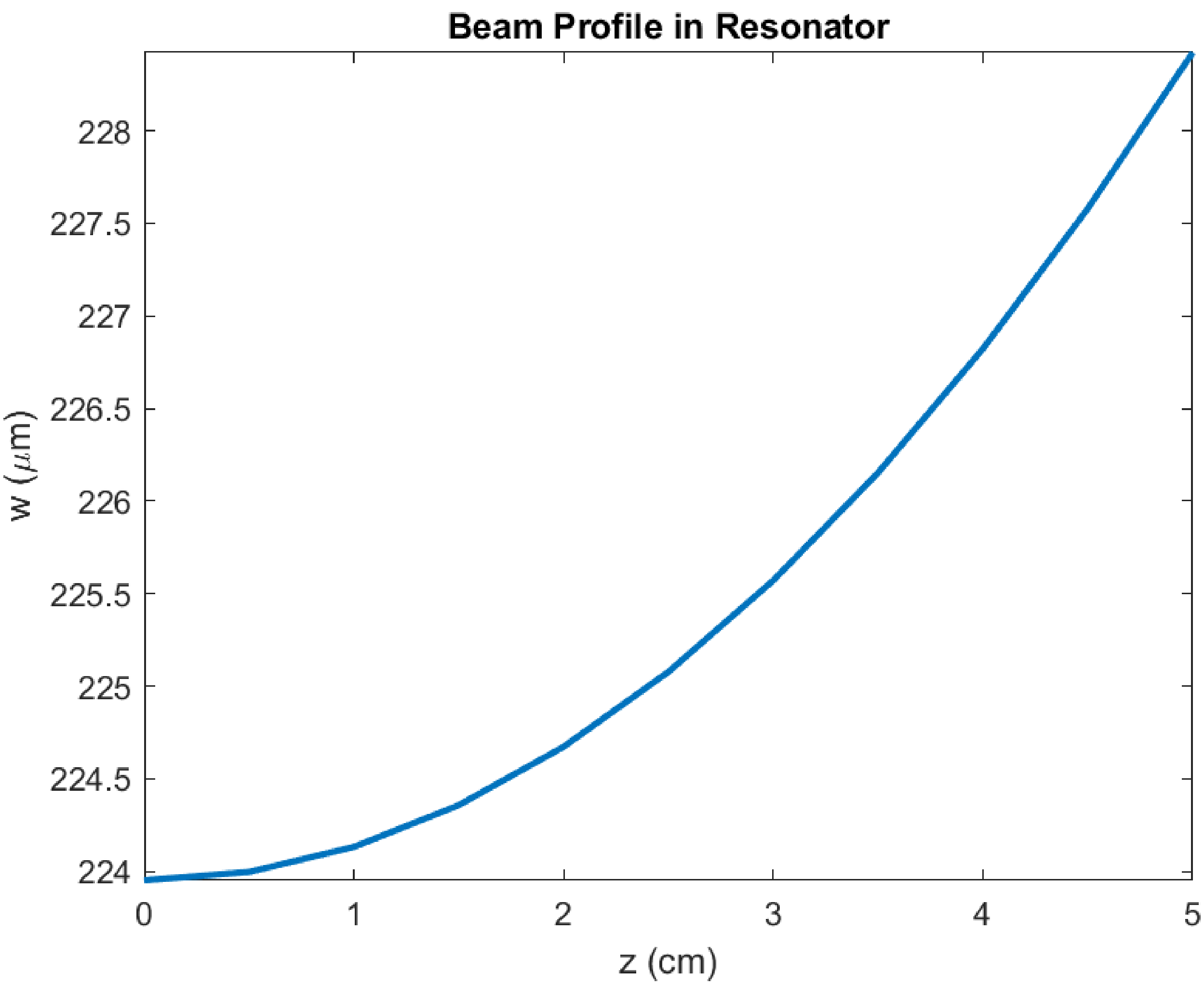
$$T = \begin{bmatrix} 1 - \frac{2(d + z_1)}{R_2} & d + z_1 + (d - z_1)\left(1 - \frac{2(d + z_1)}{R_2}\right) \\ -\frac{1}{f} & 1 - \frac{2(d - z_1)}{R_2} \end{bmatrix}$$

$$\frac{\pi\omega^2}{\lambda} = \frac{B}{\left[1 - \left(\frac{A + D}{2}\right)^2\right]^{\frac{1}{2}}}$$

In MATLAB, at $\lambda = 1063.8\ nm$, we measured the beam waist at the flat mirror $z_1 = 0$ and at the spherical mirror $z_1 = d$ separately. Here, we assumed the dimension for $d = 5\ cm$ and ranged the values for the radius of curvature for the spherical mirror from 0.06 m to 10 m in increments of 0.01 m to find the smallest radius of curvature for which the percent difference between the beam waist at each mirror is less than or equal to 2%. Starting at 0.06 m ensures we maintain the stability condition of the resonator described in the next section.

$R_2 = 1.29\ m$

The following graph showcases that the 2% uniform criteria of the beam profile is maintained for the entire length of the resonator.



We further verify that the beam waist we modeled meets the 1 cm transverse constraint as $2 * 228\ \mu m \ll 1\ cm$. Lastly, with the 10 cm constraint of the entire laser pointer, we can calculate L_2 as

$$L_2 = 10 - L_1 - d = 10 - 2.09 - 5 = 2.91\ cm$$

Thus, the complete dimensions of the pump, coupling optic, and resonator for the laser pointer are:

$\boldsymbol{L_1 = 2.09\ cm}$

$\boldsymbol{\omega_0 = 175\ \mu m}$

$\boldsymbol{d = 5\ cm}$

$\boldsymbol{R_2 = 1.29\ m}$

$\boldsymbol{L_2 = 2.91\ cm}$

$\boldsymbol{f = 70\ cm}$

$\boldsymbol{D = 1\ mm\ (diameter\ of\ thin\ lens)}$

2. Resonator Design

The resonator design implemented into the laser point is created using an optical cavity consisting of two mirrors, M_1 and M_2. These mirrors are dichroic which allows for the transmission of light at certain wavelengths. To ensure that the resonator outputs light at 532 nm and allows all green light to leave the resonator, the reflectivity of the output coupler for 532 nm is assumed to be 0. The dichroic characteristic of the mirrors allows for the cavity to have a different reflectivity at the fundamental wavelength compared to the doubled wavelength. For the back mirror M_1, the reflectivity is assumed to be 99.5% (because mirrors are realistically never truly 100% reflective) for the fundamental and 100% for the doubled wavelength which allows light to remain in the cavity and oscillate in the direction of the output coupler. We define these terms as

$$reflectivity\ at\ M_1\ @\ \lambda = 1063.8\ nm: R_1 = 0.995$$

$$reflectivity\ at\ M_2\ @\ \lambda = 532\ nm: R_2 = 0$$

The remaining unknown reflectivity is for the output coupler at the fundamental wavelength. We define this design parameter as

$$reflectivity\ at\ M_2\ @\ \lambda = 1063.8\ nm: R_2$$

There is a $Nd{:}GdVO_4$ laser gain medium located at length d_1 from the back mirror with length l_g. Located a distance d_2 from the right of the gain medium is a doubler which converts the fundamental wavelength of 1063.8 nm to 531.9 ≈ 532 nm. The doubler is located a distance of d_3 from the output coupler and has a length of 0. Therefore,

$$d = d_1 + l_g + d_2 + d_3 = 5\ cm$$

To determine the reflectivity of the output coupler at the fundamental wavelength, the radius of curvature of the output coupler, and the dimensions of the optical cavity with respect to the gain medium and doubler, we first ensured that the design of the optical cavity is stable. We have the following ABCD matrix of the optical cavity

$$T = \begin{bmatrix} 1 & d \\ -\frac{1}{f_1} & \left(1 - \frac{d}{f_1}\right) \end{bmatrix} \begin{bmatrix} 1 & d \\ -\frac{1}{f_2} & 1 - \frac{d}{f_2} \end{bmatrix} = \begin{bmatrix} 1 - \frac{d}{f_2} & d + d\left(1 - \frac{d}{f_2}\right) \\ -\frac{1}{f_1} - \frac{1}{f_2}\left(1 - \frac{d}{f_1}\right) & -\frac{d}{f_1} + \left(1 - \frac{d}{f_1}\right)\left(1 - \frac{d}{f_2}\right) \end{bmatrix}$$

For the optical cavity to be stable, the following condition must be met:

$$0 \leq \frac{A + D + 2}{4} \leq 1$$

$$A = 1 - \frac{d}{f_2}$$

$$D=-\frac{d}{f_1}+\left(1-\frac{d}{f_1}\right)\left(1-\frac{d}{f_2}\right)$$

$$A+D=1-\frac{d}{f_2}-\frac{d}{f_1}+\left(1-\frac{d}{f_1}\right)\left(1-\frac{d}{f_2}\right)$$

$$=1-\frac{d}{f_2}-\frac{d}{f_1}+1-\frac{d}{f_2}-\frac{d}{f_1}+\frac{d^2}{f_1f_2}\ =2-\frac{2d}{f_2}-\frac{2d}{f_1}+\frac{d^2}{f_1f_2}$$

$$\frac{A+D+2}{4}=\frac{2-\frac{2d}{f_2}-\frac{2d}{f_1}+\frac{d^2}{f_1f_2}+2}{4}=\frac{4-\frac{2d}{f_2}-\frac{2d}{f_1}+\frac{d^2}{f_1f_2}}{4}=1-\frac{d}{2f_2}-\frac{d}{2f_1}+\frac{d^2}{4f_1f_2}$$

$$\frac{A+D+2}{4}=\left(1-\frac{d}{2f_1}\right)\left(1-\frac{d}{2f_2}\right)$$

$$2f_1=R_1, 2f_2=R_2$$

$$\frac{A+D+2}{4}=\left(1-\frac{d}{R_1}\right)\left(1-\frac{d}{R_2}\right)$$

$$R_1=\infty$$

$$\boldsymbol{0\leq\left(1-\frac{d}{R_2}\right)\leq 1}$$

We previously calculated our cavity length and mirror radius of curvature for the pump dimensions (which will be used for further calculations). Thus, we verify

$$\left(1-\frac{0.05}{1.29}\right)=0.96$$

$$0\leq 0.96\leq 1$$

With the inclusion of a gain medium, mirrors with non-perfect reflectivities, and a doubler, the beam is known to experience both gain and loss through the resonator. To clarify, the 808.5 nm light pumps the gain medium to excite atoms and release photons at 1063.8 nm which then pass through the doubler to produce a 532 nm wavelength. The gain and loss experienced in the resonator is applied only to the 1063.8 nm beam. Thus, we consider, the resulting loss and gain for a total round trip, passing through the gain medium of length l_g and doubler twice to be

$$R_1R_2e^{2\sigma\Delta N_0 l_g}e^{-2l_gL}=1$$

where L is the coupling efficiency of the doubler used to determine the intensity-dependent loss contributing to the lasing threshold at the fundamental wavelength.

$$L=\tanh^2\sqrt{I}$$

Here, the higher coupling efficiency, the more loss experienced by the 1063.8 nm beam, resulting in an additional loss term of $2L$ due to the round trip of the fundamental mode passing through the doubler twice. Using the equations from above, we can determine what our threshold gain is

$$\gamma_{th} = \Delta N_0 \sigma = 2L + \frac{1}{2l_g} \ln \frac{1}{R_1 R_2}$$

For the design of this resonator and laser pointer, we use the gain calculated in the previous section that allowed for a 1 mW output green laser. Before moving forward, we consider gain saturation.

$$I_s = \frac{hv}{\sigma \tau_2} = \frac{6.626 * 10^{-34} * 3 * 10^{8}}{1063.8 * 10^{-9}} * \frac{1}{7.6 * 10^{-19} * 100 * 10^{-6}} = 2.4587\ kW/cm^2$$

Here, since $I_1 < I_s$, we ignore gain saturation and move forward $G_0 = \frac{I_2}{I_1}$.

$$G_0 = e^{\gamma_0 l_g}$$

$$\ln G_0 = \gamma_0 l_g$$

$$\frac{\ln 24}{l_g} = \gamma_0$$

and start with the assumption that $l_g = 4\ cm$ and $L = 0.0000449072$ from our previous calculation.

$$\gamma_0 = \frac{\ln 21}{4} = 0.7945\ cm^{-1}$$

$$I = \left(\tanh^{-1} \sqrt{0.0000449072}\right)^2 = 0.0000449085\ GW/cm^2$$

$$\gamma_{th} = \frac{1}{8} \ln\left(\frac{1}{0.995 R_2}\right) + 2 * 0.0000449072$$

From Section 9.2.3(a), we use the analysis of a standing wave laser with high-Q approximation to determine the value of the transmission coefficient of M_2 that yields the maximum output intensity.

$$I_{out} = \frac{T_b I_s}{2} \left[T_2 \left(\frac{g_0}{L' + T} - 1\right)\right]$$

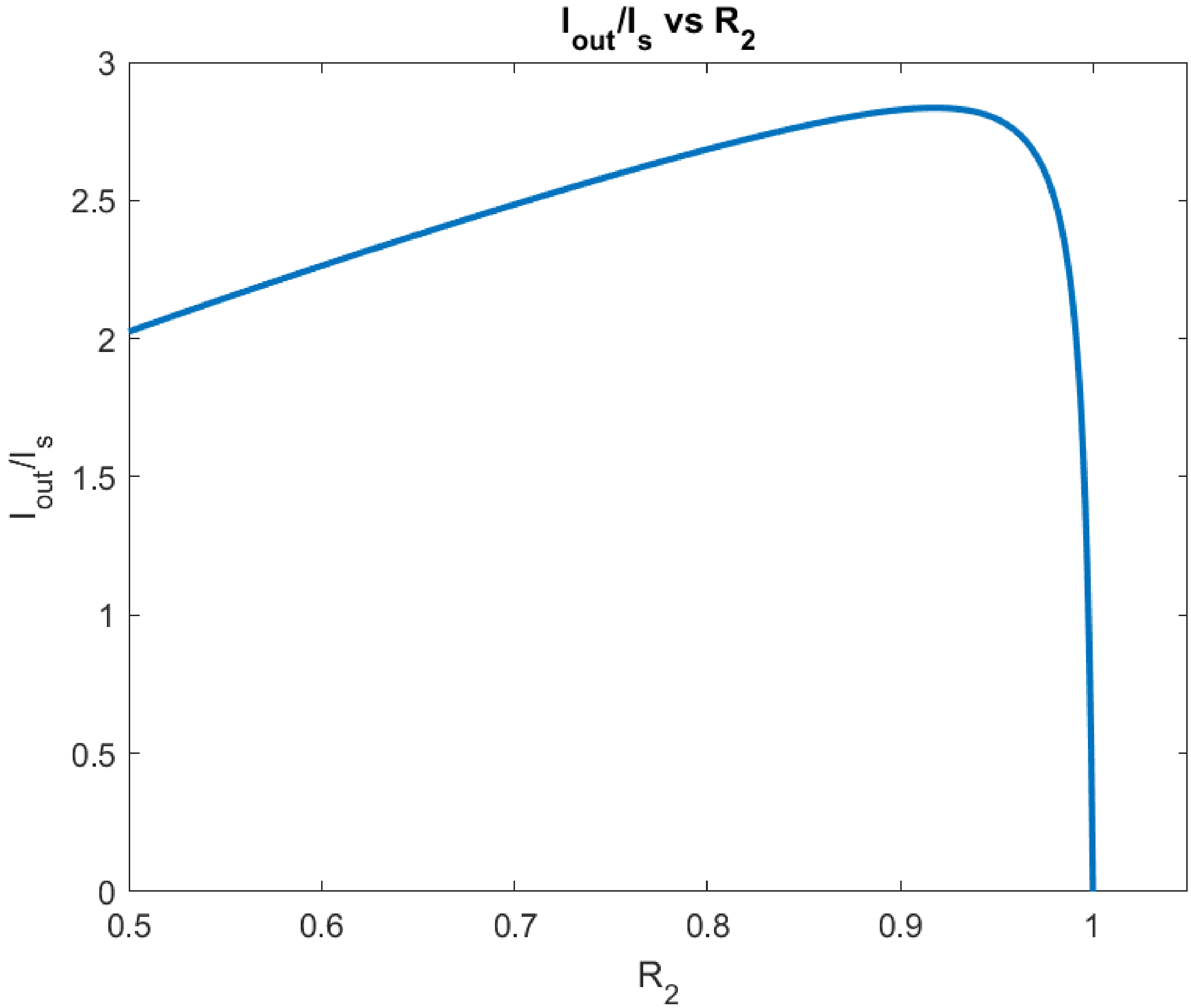
We assume that $T_b = 1$ and define $g_0 = 2\gamma_0 l_g$, $T_2 = 1 - R_2$ while $L + T$ represents the total loss as

$$L' + T = \ln\left(\frac{1}{R_1 R_2}\right) + L$$

In the output intensity equation, we know that I_s is a constant and thus redefine the equation to be

$$\frac{I_{out}}{I_s} = \frac{1}{2}\left[(1 - R_2)\left(\frac{2\gamma_0(v) l_g}{\ln\left(\frac{1}{R_2}\right) + L} - 1\right)\right]$$

We now have a function in terms of I_{out} and R_2 and plot the function to determine at what reflectivity we receive the maximum output intensity.



Here, the maximum output intensity is achieved when $R_2 = 0.918$

$$\gamma_{th} = \frac{1}{8}\ln\left(\frac{1}{0.995 * 0.918}\right) + 2 * 0.0000449072 = 0.011411$$

From the graph, we retrieve the value of $\frac{I_{out}}{I_s}$ where $R_2 = 0.918$.

$$\frac{I_{out}}{I_s} = 2.835$$

$$I_{out} = 2.835 * I_s = 6.97\ kW/cm^2$$

We will verify that our assumption for a high-Q approximation is correct.

$$Q = \frac{\omega}{\Delta\omega}$$

$$\omega = \frac{2\pi c}{\lambda} = \frac{2\pi(3 * 10^8)}{(1063.8 * 10^{-9})} = 1.772 * 10^{15}$$

$$\Delta\omega\tau_p = 1$$

$$\tau_p = \frac{\frac{2d}{c}}{1 - R_1R_2}$$

$$\tau_p = \frac{\frac{2(0.05)}{3 * 10^8}}{1 - 0.995 * 0.918} = 3.85 * 10^{-9}$$

$$Q = \frac{1.772 * 10^{15}}{3.85 * 10^{-9}} = 4.603 * 10^{23}$$

The maximum output intensity from the high-Q approximation formula used in Section 9.2.3(a) decreases as the length of the gain medium decreases, assuming we keep the intensity that causes the intensity-dependent loss constant. Furthermore, since the beam profile was assumed uniform, the distance between the gain medium and the doubler and mirrors is trivial. Thus, with an assumed $l_g = 4\ cm$ and an assumed $d = 5\ cm$, we center the gain medium in the resonator resulting in $d_2 = 0.25\ cm, d_3 = 0.25\ cm, d_1 = 0.5\ cm$.

$$d = d_1 + l_g + d_2 + d_3$$

$$5 = d_1 + 4 + 0.25 + 0.25$$

$$d_1 = 5 - 4 - 0.5 = 0.5\ cm$$

We are asked to also find the cross-sectional area A of the gain medium. This is determined in the section with rate equations that allows for sufficient pump power.

$$A = 1.3056 * 10^{-6}\ cm^2$$

Our dimensions for the resonator are then

$\boldsymbol{R_1 = 0.995}$

$\boldsymbol{R_2 = 0.918}$

$\boldsymbol{d_1 = 0.5\ cm}$

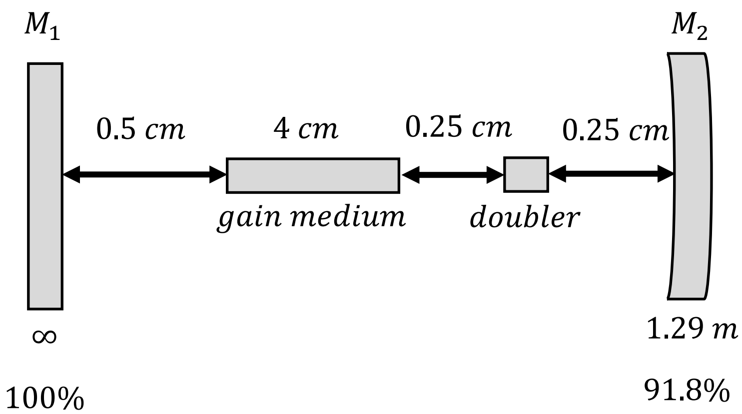
$\boldsymbol{d_2 = 0.25\ cm}$

$\boldsymbol{d_3 = 0.25\ cm}$

$\boldsymbol{l_g = 4\ cm}$

$\boldsymbol{A = 1.3056 * 10^{-6}\ cm^2}$

The following figure is a sketch depicting the location of the gain medium and doubler with all the distances in the optical cavity.



3. Rate Equations

For the rate equations, we calculate with the knowledge that the Nd:GdVO$_4$ creates an initial population in the upper laser state by the pumping agent with negligible population in the pumped states. Thus, $N_3 \sim 0$ and every atom pumped to state 3 relaxes to state 2. We first calculate what the threshold population inversion needs to be to achieve oscillation. We also assume that there is no depletion of state 0.

$$\gamma_{th} = \sigma(v) N_{th}$$

$$\frac{\gamma_{th}}{\sigma(v)} = N_{th}$$

$$\gamma_{th} = 0.011411$$

$$\frac{0.011411}{7.6 * 10^{-19}} = N_{th}$$

$$N_{th} = 1.5015 * 10^{16}\ cm^{-3}$$

$$\sigma_{abs}(v) = \frac{g_2}{g_1} \sigma_{stim}(v)$$

$$\sigma_{abs}(v) = \frac{g_2}{g_1}(7.6 * 10^{-19})$$

$$5.1 * 10^{-19} = \frac{g_2}{g_1}(7.6 * 10^{-19})$$

$$\frac{g_2}{g_1} = \frac{5.1 * 10^{-19}}{7.6 * 10^{-19}} = \frac{5.1}{7.6}$$

$$\Delta N_0 = N_{th}$$

(1) $\Delta N_0 = N_2 - \frac{g_2}{g_1} N_1$

(2) $N_2 + N_1 = Nd$

$N_1 = Nd - N_2$

$$\Delta N_0 = N_2 - \frac{g_2}{g_1}(Nd - N_2)$$

$$\Delta N_0 = N_2 - \frac{g_2}{g_1} Nd + \frac{g_2}{g_1} N_2$$

$$\Delta N_0 = N_2 \left(1 + \frac{g_2}{g_1}\right) - \frac{g_2}{g_1} Nd$$

$$\frac{\Delta N_0 + \frac{g_2}{g_1} Nd}{\left(1 + \frac{g_2}{g_1}\right)} = N_2$$

$N_2 = 5.62 * 10^{19}\ cm^{-3}$

$N_1 = Nd - N_2 = 1.4 * 10^{20} - 5.62 * 10^{19} = 8.377 * 10^{19}\ cm^{-3}$

Note here that $N_1 > N_2$ but the difference in the stimulated and absorption cross sectional area shows there is a degeneracy rate, and we also have gain. Knowing what the threshold population inversion and the population in the upper state would be, we can determine what the pump rate is to reach the population inversion.

$$\frac{dN_2}{dt} = P - \frac{N_2}{\tau_{21}} - \frac{\sigma_{21} I_1}{h v_{21}} \Delta N_0 = 0$$

Here, I_1 is also referred to as the pump intensity I_p.

$$P = \frac{N_2}{\tau_{21}} + \frac{\sigma_{21} I_1}{h v_{21}} \Delta N_0$$

$$P = \frac{5.62 * 10^{19}}{100 * 10^{-6}} + \frac{7.6 * 10^{-19} * 1916}{\frac{6.626 * 10^{-34} * 3 * 10^{8}}{1063.8 * 10^{-9}}} 1.5015 * 10^{16} = 5.624 * 10^{23}\ cm^{-3} s^{-1}$$

$$\frac{Power}{Volume} = 1.165 * 1.6022 * 10^{-19} * P = 104.97 \frac{kW}{cm^3}$$

We can use the relationship between the pump power needed to maintain the population at state 2 and the volume of the gain medium to determine what the cross-sectional area A needs to be for the gain medium.

$$P_{pump} = \frac{\lambda_1}{\lambda_3 \eta_{qe}} P_{spont}$$

$$P_{spont} = \frac{N_2 V h v}{\tau_2}$$

$$P_{pump} = \frac{\lambda_1}{\lambda_3 \eta_{qe}} \frac{N_2 V h v}{\tau_2}$$

$$\frac{P_{pump} \lambda_3 \eta_{qe} \tau_2}{\lambda_1 N_2 h v} = V$$

$$\frac{0.95 * 808.5 * 10^{-9} * 0.76 * 100 * 10^{-6} * 1063.8 * 10^{-9}}{1063.8 * 10^{-9} * 5.62 * 10^{19} * 6.626 * 10^{-34} * 3 * 10^{8}} = V$$

$$V = 5.22 * 10^{-6}\ cm^3$$

$$V = A l_g$$

$$\frac{V}{l_g} = A$$

$$\frac{5.22 * 10^{-6}}{4} = A$$

$$\boldsymbol{A = 1.3056 * 10^{-6}\ cm^2}$$

Thus, we are satisfied with our design parameters thus far.

4. Pump Absorption

Now, with the volume of the gain medium provided, we can check whether our $0.95\ W$ pump power meets the threshold pump required as stated by our rate equations across the gain medium.

$$\frac{P_{pump}}{V} = \frac{0.95}{5.22*10^{-6}} = 181.9 \frac{kW}{cm^3} > 104.97 \frac{kW}{cm^3}$$

We define the pump absorption to be

$$P_{abs} = I_p(0) A \left(1 - e^{-\gamma_{th} l_g}\right)$$

where $I_p(0) = I_1$ and A is the area of the pump beam.

$$A = \frac{\pi \omega^2}{2} = \frac{\pi (178 * 10^{-4})^2}{2} = 4.9583 * 10^{-4}$$

$$P_{abs} = 1916 * 4.9583 * 10^{-4} (1 - e^{-0.0456}) = \mathbf{0.0424\ W}$$

5. Output power and efficiency

The calculations from previous sections were performed with the goal of designing a laser pointer to output 1 mW of power for the green wavelength. Here, we will verify that our design results in this output power.

$$I_0 = \frac{2P_0}{\pi \omega_0^2} = \frac{2(1)}{\pi (175 * 10^{-4})^2} = 2078.8 \frac{W}{cm^2}$$

$$I(r,z) = I_0 \left(\frac{\omega_0}{\omega(z)}\right)^2 \left[e^{-\frac{2r^2}{\omega^2(z)}}\right]$$

$$I(0,L_1) = 2078.8 \left(\frac{175}{178}\right)^2 [e^0] = 2016.8 \frac{W}{cm^2}$$

$$I_1 = 0.95 * I(0,L_1) = 0.95 * 2016.8 = 1916 \frac{W}{cm^2}$$

$$P_1 = I_1 A_{beam} = 1916 * \frac{\pi(178 * 10^{-4})^2}{2} = 0.95\ W$$

$$I_s = \frac{hv}{\sigma\tau_2} = \frac{6.626 * 10^{-34} * 3 * 10^8}{1063.8 * 10^{-9}} * \frac{1}{7.6 * 10^{-19} * 100 * 10^{-6}} = 2.4587\ kW/cm^2$$

Since $I_1 < I_s$, we can ignore gain saturation.

$$G_0 = \frac{P_2}{P_1} = 24$$

$$\frac{P_2}{0.95} = 24$$

$$P_2 = 24 * 0.95 = 22.8\ W$$

$$I_2 = \frac{2P_2}{\pi\omega_0^2} = \frac{2 * 22.8}{\pi(178 * 10^{-4})^2} = 45.8\ kW/cm^2$$

$$L = \tanh^2 \sqrt{I}$$

$$L = \tanh^2 \sqrt{0.000045812} = 0.0000458106$$

$$I_{out} = 45.8 * L = 2.098\ W/cm^2$$

$$P_{out} = 2.098 * \pi * \frac{\omega_0^2}{2} = 2.098 * \pi * \frac{(178 * 10^{-4})^2}{2} * 1000 = \mathbf{1.04\ mW} \ \approx \mathbf{1\ mW}$$

Now, we analyze the efficiency of our laser pointer design.

$$\eta_{total} = \eta_{qe} * \eta_{cpl} * \eta_{pump}$$

$$\eta_{cpl} = \frac{T_b T_2}{\left(1 - \sqrt{T_a^2 T_b^2 R_1 R_2}\right)\left(1 - \sqrt{T_a^2 T_b^2 R_1 R_2}\right)} = \frac{1 - R_2}{\left(1 - \sqrt{R_1 R_2}\right)\left(1 - \sqrt{R_1 R_2}\right)}$$

$$\eta_{cpl} = \frac{1 - 0.918}{\left(1 - \sqrt{0.995 * 0.918}\right)\left(1 - \sqrt{0.995 * 0.918}\right)} = 0.947$$

We assume a simple rate equation for the upper state and recognize that lasers have the upper state clamped at the threshold value.

$$\frac{dN_2(z)}{dt} = \eta_p\left(\frac{\sigma_{21}I_p^{th}(z)}{hv_p}\right)\Delta N_0 - \frac{N_2}{\tau_2} = 0$$

Here, we assume the pump decreases exponentially with z, but we operate in steady state. Thus,

$$0 = \eta_p\left(\frac{\sigma_{21}I_p^{th}(z)}{hv_p}\right)\Delta N_0 - \frac{N_2}{\tau_2}$$

$$N_2(0) = \eta_p\left(\frac{\sigma_{21}I_p^{th}(0)}{hv_p}\right)\Delta N_0\tau_2$$

$$N_2(0)\sigma_{21} = \eta_p\left(\frac{\sigma_{21}\tau_2}{hv_p}\right)I_p^{th}(0)\Delta N_0\sigma_{21}$$

$$\gamma_{th} = N_2(0)\sigma_{21} = \eta_p\left(\frac{\sigma_{21}\tau_2}{hv_{21}}\right)\left(\frac{hv_{21}}{hv_p}\right)I_p^{th}(0)(\Delta N_0\sigma_{21})$$

$$\left(\frac{hv_{21}}{hv_p}\right) = \eta_{qe}$$

$$hv_{21} = \frac{6.626 * 10^{-34} * 3 * 10^8}{1063.8 * 10^{-9}} = 1.86858 * 10^{-19}$$

$$hv_p = \frac{6.626 * 10^{-34} * 3 * 10^8}{808.5 * 10^{-9}} = 2.45862 * 10^{-19}$$

$$\eta_{qe} = \frac{hv_{21}}{hv_p} = \frac{1.86858 * 10^{-19}}{2.45862 * 10^{-19}} = 0.76$$

$$\eta_p = \frac{P_{abs}}{P_1} = \frac{0.424}{0.95} = 0.0446$$

$$\eta = 0.0446 * 0.947 * 0.76 = 0.0321 \equiv \mathbf{3.21}\%$$

6. MATLAB Code

```
clear
% Coupling Optics
w0 = 175e-6;
lambda1 = 808.5e-9;
lambda2 = 1063.8e-9;
z0 = pi*w0^2/lambda1;
f = 0:0.1:1;
L1 = zeros(1, length(f));
syms d
```

```
for i = 1:length(f)
    eqn = d^2-d*f(i)+z0^2 == 0;
    if ~isempty(solve(eqn, d))
        L1(i) = min(double(solve(eqn, d)));
        if imag(L1(i)) ~= 0
            L1(i) = 0;
        end
    else
        L1(i) = NaN;
    end
end
f(L1 == 0 | L1 < 0.02 | L1 > 0.05) = [];
L1(L1 == 0 | L1 < 0.02 | L1 > 0.05) = [];
f = f(L1 == min(L1));
L1 = min(L1);
wz = w0*sqrt(1+(L1/z0)^2);
diff = (wz-w0)/w0 * 100
w0 = w0*100;
wz = wz*100;
% Conversion Efficiency and Gain
P_out = 1;
syms P2
I = sqrt((2*P2*10^-9)/(pi*wz^2));
eqn = P2 * 10^-9 * tanh(I)^2 * 10^12 == P_out;
P2 = double(abs(vpasolve(eqn, P2)))
I = sqrt((2*P2*10^-9)/(pi*wz^2));
L = tanh(I)^2
P0 = 1;
I0 = 2*P0/(pi*w0^2)
Irz = I0*(w0/wz)^2
P1 = 0.95*Irz*pi*wz^2/2
G = ceil(P2 / P1);
% M2 reflectivity and output intensity
lg = 4;
I1 = P1*2/(wz^2*pi);
R1 = 0.995;
R2 = 0.5:0.001:1;
sigmaST = 7.6e-19;
sigmaABS = 5.1e-19;
tau21 = 100e-6;
F = log(1./(R1*R2)) + L;
T2 = 1-R2;
g0 = log(G)/lg;
I2 = (atanh(sqrt(L)))^2
I_r = 1/2*(T2.*((2*g0*lg./F)-1));
figure
plot(R2, I_r, 'LineWidth', 2)
xlabel('R_{2}')
ylabel('I_{out}/I_{s}')
title('I_{out}/I_{s} vs R_{2}')
```

```
axis tight
ylim([0, 3])
xlim([0.5, 1.05])
R2 = R2(I_r == max(I_r))
at=1/(2*lg)*log(1/(R1*R2));
gth = at + 2*L;
Is = (6.626e-34*3e8/lambda2)/(sigmaST*tau21)
I_o = max(I_r)*Is
% Finding R2 (radius)
radius2 = 0.06:0.01:10;
d = 0.05;
lambda2 = 6328e-10;
for i = 1:length(radius2)
    z1 = 0;
    T = [1 d+z1; 0 1]*[1 0; -2./radius2(i) 1]*[1 d-z1; 0 1];
    B = T(1,2);
    A = T(1,1);
    D = T(2,2);
    w_m1 = sqrt(lambda2*B/(pi*sqrt(1-((A+D)/2)^2)));
    z1 = d;
    T = [1 d+z1; 0 1]*[1 0; -2./radius2(i) 1]*[1 d-z1; 0 1];
    B = T(1,2);
    A = T(1,1);
    D = T(2,2);
    w_m2 = sqrt(lambda2*B/(pi*sqrt(1-((A+D)/2)^2)));
    diff_w = (w_m2 - w_m1)*100/w_m1;
    if diff_w <= 2
        radius2 = radius2(i)
        break
    end
end
z1 = 0:0.005:0.05;
w_m = zeros(1, length(z1));
for i = 1:length(z1)
    T = [1 d+z1(i); 0 1]*[1 0; -2./radius2 1]*[1 d-z1(i); 0 1];
    B = T(1,2);
    A = T(1,1);
    D = T(2,2);
    w_m(i) = sqrt(lambda2*B/(pi*sqrt(1-((A+D)/2)^2)));
end
plot(z1*100, w_m*10^6, 'LineWidth', 2)
xlabel('z (cm)')
ylabel('w ({\mu}m)')
title('Beam Profile in Resonator')
axis tight
% Rate Equations and Efficiency
deltaN = gth/sigmaST
g2g1 = sigmaABS/sigmaST
Nd = 1.4e20;
N2 = (deltaN + g2g1*Nd)/(1+g2g1)
```

```
N1 = Nd - N2
hv = (6.626e-34*3e8/lambda2);
P = N2/tau21 + sigmaST*I1*deltaN/hv
pp = (1.165)*1.6022e-19*P / 1000
V = P1*lambda1*0.76*tau21/(lambda2*N2*hv)
ppump = P1/(V*1000)
A = V/lg
Ap = pi*wz^2/2
Pabs = I1*Ap*(1-exp(-gth*lg))
np = Pabs/0.95
n = np*0.76*0.947
```